Crime fell during one of the worst economic shocks in modern history.

Mobility, Economic Conditions, and Crime During COVID-19

Evidence from Boston | 2018 to 2022

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Why? Did mobility restrictions or unemployment drive the fall in crime?

Introduction & Research Question

Context

COVID-19 was a dual shock:

- Labor market: unemployment 2.9% → 16.5%
- Mobility: transit down ~63% from baseline

Classic models (Becker 1968) predict higher unemployment → more crime.

But crime actually fell. Why?

Research Question

Did mobility restrictions outweigh economic distress in shaping Boston crime during COVID-19?

Framework: Routine Activity Theory
(Cohen & Felson 1979): crime requires convergence of offender, target, and lack of guardianship in space and time.

Data: BPD incidents · FRED labor market · Google Mobility · Wikipedia scrape

N = 60 monthly observations, Boston 2018–2022

Summary Statistics

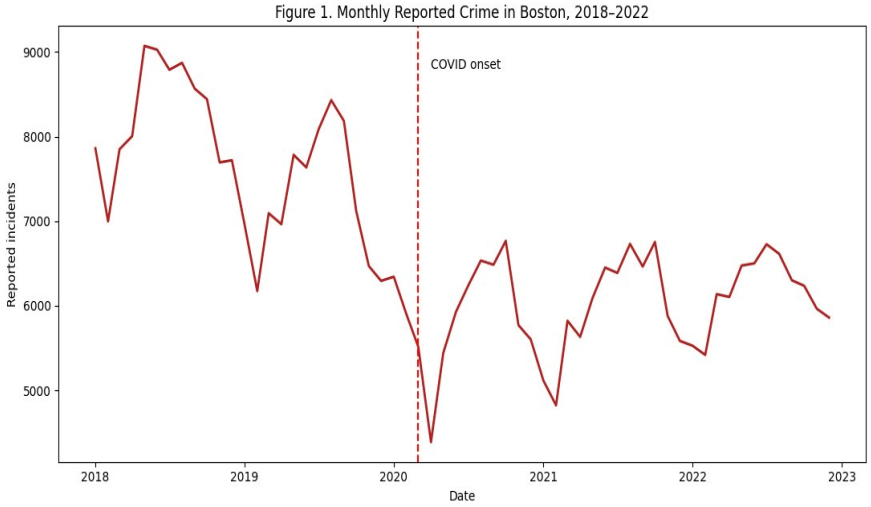
Period	Avg Monthly Crime	Unemployment Rate	Labor Force	Transit Mobility
Pre-COVID	**7,627**	2.92%	2,839,844	≈ Baseline
COVID Shock (2020)	**5,866 ↓**	**10.23% ↑**	2,770,492	**−62.5%**
Post-COVID (2021–22)	**6,066**	4.12%	2,777,711	N/A

Crime fell 23% during the COVID Shock even as unemployment tripled, the opposite of classic economic-distress predictions.

N = 60 monthly observations | Transit mobility available for 11 months in 2020 only

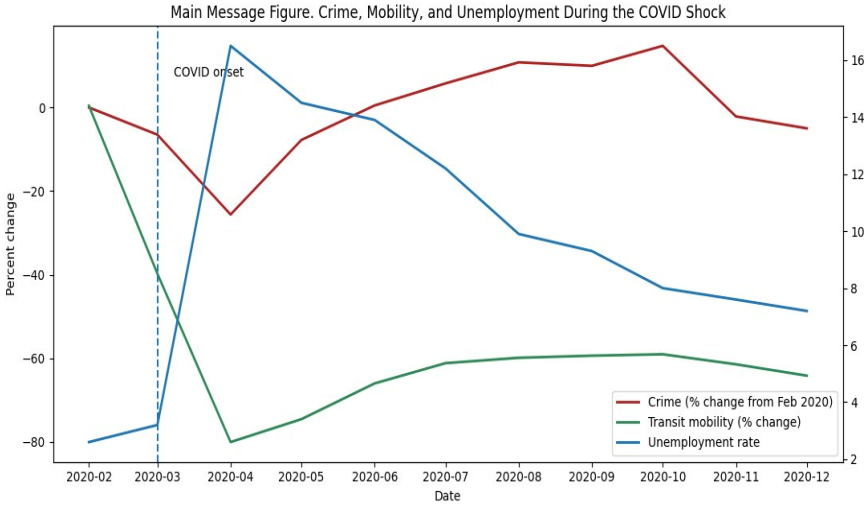
Key Visualizations

Figure 1: Monthly Reported Crime, Boston 2018–2022



A sharp structural break appears at the onset of COVID-19 in March 2020. The decline is much larger than normal seasonal variation, and crime only partially recovers afterward.

Main Message: Crime, Mobility & Unemployment (2020)



At the start of lockdowns, crime and transit mobility fall together while unemployment spikes sharply. This pattern supports the idea that reduced movement through the city played a key role in the short-run decline in crime.

Regression Results

Preferred Specification (Model 3): total_crime ~ unemployment_rate + covid + trend + labor_force | Adj. $R^2 = 0.585$

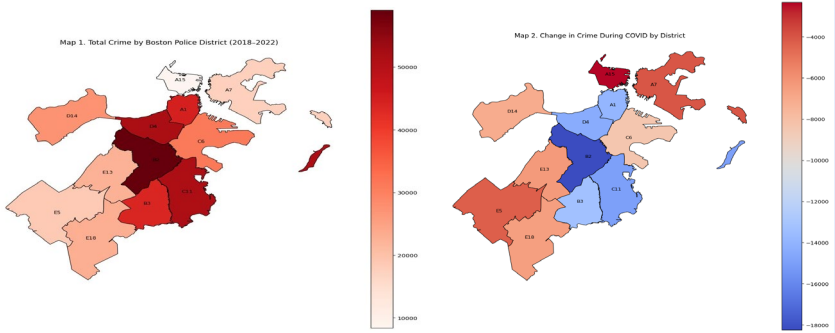
Variable	Coeff. (M1)	Coeff. (M3 ★)	Sig.	Interpretation
Unemployment Rate (%)	−117.9	−99.9	**	Higher unemployment → lower crime (contradicts distress model)
COVID Period Dummy	−311	+29	n.s.	Effect absorbed by time trend once trend is controlled
Time Trend	−31.9	−34.6	**	Crime declining ~32–35 incidents/month over sample period
Transit Mobility (M6)	—	**+128.2**	**	Higher mobility → more crime; supports Routine Activity Theory

★ Preferred spec highest adj. R^2 | ** $p < 0.05$ | Mobility models (M5–M8) use 11-obs sample; directional evidence only

Maps, Web Scraping & Machine Learning

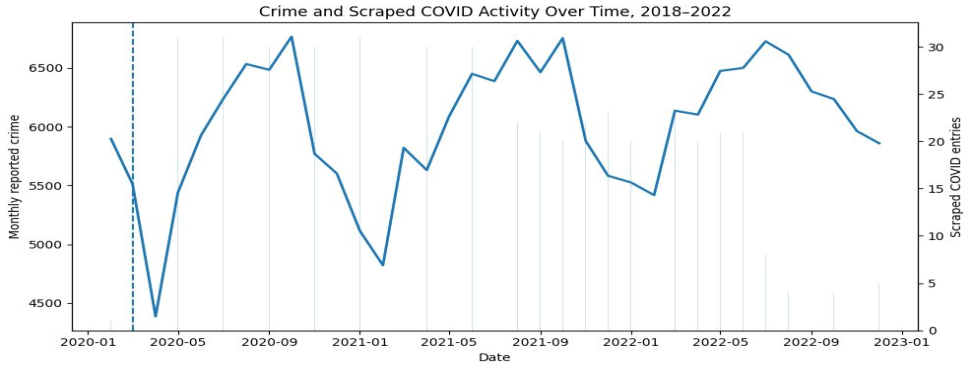
Crime concentrated in central districts (B2, D4, A1). Not uniform across the city.

Map 1 and 2: Crime and Changes in Crime by District

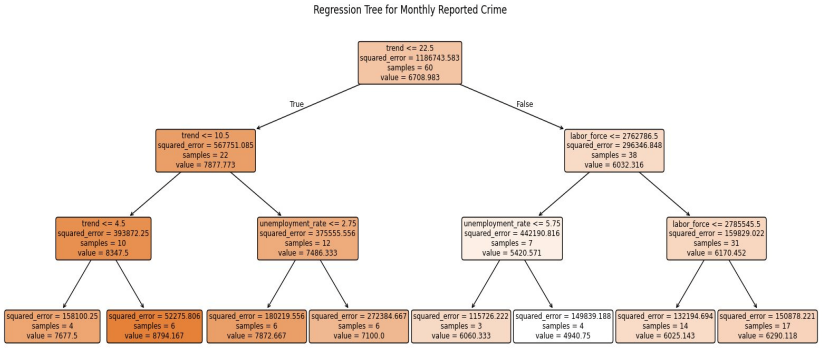


Peak COVID Wikipedia activity aligns with the sharpest crime declines, corroborates timing of the shock.

Web Scraping: Crime vs. Scraped COVID Activity (Wikipedia)

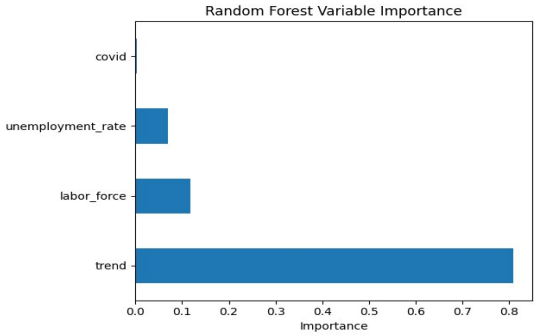


Regression Tree, first split on time trend (nonlinear, time-driven)



Both machine learning models point to the same result. The regression tree splits first on the time trend, while the random forest assigns about 81% of total importance to that same variable. Labor force and unemployment still matter, but the COVID dummy contributes almost nothing on its own. Together, these results suggest that crime changed through broader structural shifts over time rather than through a simple before-and-after COVID effect.

RF Importance: Trend 81%, COVID ≈ 0



Conclusion

Crime Fell Despite Rising Unemployment

Monthly incidents fell from 7,627 → 5,866 (−23%) while unemployment tripled to 10.2%. Classic economic-distress models cannot explain this.

Mobility Was the Key Mechanism

Transit mobility collapsed 63% alongside crime. Regression shows mobility positively predicts crime (coeff. +128.2, $p<0.05$), consistent with Routine Activity Theory.

Time Trend Dominates ML Models

Random forest assigns 81% importance to the time trend; COVID dummy ≈ 0. Pandemic mattered through structural changes, not as a binary event.

Limitations: Mobility data limited to 2020 only. Next steps: longer mobility series, offense-specific models, policy stringency index.